13. Courses of Study and Scheme of Assessment

BE INSTRUMENTATION AND CONTROL ENGINEERING (2023 REGULATIONS)

(Minimum No. of credits to be earned: 166)

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER I										
THEORY										
1	23U101	Calculus and its Applications	3	1	0	4	40	60	100	BS
2	23U102	Physics	3	0	0	3	40	60	100	BS
3	23U103	Electric Circuits	3	1	0	4	40	60	100	ES
4	23U104	Electronic Devices and Circuits	3	0	0	3	40	60	100	ES
5	23G105	English Language Proficiency	3	1	0	4	40	60	100	HS
PRACTICALS										
6	23U110	Engineering Graphics	0	0	4	2	60	40	100	ES
7	23U111	Circuits and Devices Laboratory	0	0	4	2	60	40	100	ES
MANDATORY COURSES										
8	23IP15	Induction Programme	-	-	-	Grade	-	-		MC
		Total 26hrs	**15**	**3**	**8**	**22**	**320**	**380**	**700**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER II										
THEORY										
1	23U201	Complex Variables and Transforms	3	1	0	4	40	60	100	BS
2	23U202	Applied Chemistry	3	0	0	3	40	60	100	BS
3	23U203	Digital Electronics	3	1	0	4	40	60	100	ES
4	23U204	Linear Integrated Circuits	3	0	0	3	40	60	100	ES
PRACTICALS										
5	23G___	Language Elective	0	0	4	2	60	40	100	HS
6	23U210	Analog and Digital Electronics Laboratory	0	0	4	2	60	40	100	ES
7	23U211	Basic Science Laboratory	0	0	4	2	60	40	100	BS
MANDATORY COURSES										
8	23Q213	Foundations of Problem Solving	0	0	2	0	60	40	100	MC
9	23U215	Activity Point Programme *	-	-	-	Grade	-	-		MC
		Total 28hrs	**12**	**2**	**14**	**20**	**400**	**400**	**800**	

** As per norms

* As per AICTE norms; Total 60 hrs; Grade: Completed/Not Completed; Not counted for CGPA

CA Continuous Assessment

FE Final Examination

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S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER III										
THEORY										
1	23U301	Linear Algebra	3	1	0	4	40	60	100	BS
2	23U302	Sensors and Transducers	4	0	0	4	40	60	100	PC
3	23U303	C Programming	2	0	0	2	40	60	100	ES
4	23U304	Electrical Machines	4	0	0	4	40	60	100	BS
5	23O305	Engineering Economics	3	1	0	4	40	60	100	HS
PRACTICALS										
6	23U310	Sensors and Transducers Laboratory	0	0	4	2	60	40	100	PC
7	23U311	C Programming Laboratory	0	0	4	2	60	40	100	ES
8	23Q313	Building Communication Skills	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23K312	Environmental Science	2	0	0	0	-	-	-	MC
10	23U315	Activity Point Programme*	-	-	-	Grade	-	-	-	MC
11	23TC01	Heritage of Tamils / **தமிழர்மரபு**	1	0	0	1	100	0	100	HS
		Total 31 hrs	**19**	**2**	**10**	**24**	**480**	**420**	**900**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER IV										
THEORY										
1	23U401	Probability, Stochastic Processes and Statistics	3	1	0	4	40	60	100	BS
2	23U402	Electrical and Electronic Measurements	3	0	0	3	40	60	100	PC
3	23U403	Control Systems	3	1	0	4	40	60	100	PC
4	23U404	Microprocessors and Microcontrollers	4	0	0	4	40	60	100	PC
5	23U405	Data Structures and Algorithms	2	2	0	4	40	60	100	ES
PRACTICALS										
6	23U410	Microprocessors and Microcontrollers Laboratory	0	0	4	2	60	40	100	PC
7	23U411	Measurement and Control Laboratory	0	0	4	2	60	40	100	PC
8	23Q413	Problem Solving	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23O414	Indian Constitution**	2	0	0	0	-	-	-	MC
10	23U415	Activity Point Programme*	-	-	-	Grade	-	-	-	MC
11	23TC02	Tamils and Technology / தமிழரும்தொழில்நுட்பமும்	1	0	0	1	100	0	100	HS
		Total 32 hrs	**18**	**4**	**10**	**25**	**480**	**420**	**900**	

** As per norms
* As per AICTE norms; Total 60 hrs; Grade: Completed/Not Completed; Not counted for CGPA
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S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER V										
THEORY										
1	23U501	Industrial Instrumentation	3	0	0	3	40	60	100	PC
2	23U502	Digital Signal Processing	3	0	0	3	40	60	100	PC
3	23U503	Digital Control Systems	3	1	0	4	40	60	100	PC
4	23U504	Embedded System Design	3	0	0	3	40	60	100	PC
5	23U___	Professional Elective I	3	0	0	3	40	60	100	PE
PRACTICALS										
6	23U510	Signal Processing and Digital Control Laboratory	0	0	4	2	60	40	100	PC
7	23U511	Embedded Systems Laboratory	0	0	4	2	60	40	100	PC
8	23Q513	Aptitude Skills	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23U515	Activity Point Programme *	-	-	-	Grade	-	-	-	MC
	Total 26hrs		**15**	**1**	**10**	**21**	**380**	**420**	**800**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER VI										
THEORY										
1	23U601	Instrumentation System Design	4	0	0	4	40	60	100	PC
2	23U602	Industrial Process Control	3	1	0	4	40	60	100	PC
3	23U603	Artificial Intelligence for Measurement and Control	3	0	0	3	40	60	100	PC
4	23U___	Professional Elective II	3	0	0	3	40	60	100	PE
5	23____	Open Elective I	3	0	0	3	40	60	100	OE
PRACTICALS										
6	23U610	Instrumentation System Design Laboratory	0	0	4	2	60	40	100	PC
7	23U611	Process Control Laboratory	0	0	4	2	60	40	100	PC
8	23U620	Innovation Practices	0	0	2	1	60	40	100	EEC
9	23Q613	Enhancing Problem Solving Ability with Code	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
10	23U615	Activity Point Programme*	-	-	-	Grade	-	-	-	MC
	Total 29hrs		**16**	**1**	**12**	**23**	**440**	**460**	**900**	

At the end of 6th semester, the students are required to earn the minimum number of activity points from the AICTE mandated ACTIVITY POINT PROGRAMME to qualify for the award of BE/BTech degree (Refer section 4 (vii) (c) of 2023 Regulations)

* As per AICTE norms; Total 60 hrs; Grade: Completed/Not Completed; Not counted for CGPA

Grade Competed/Not Completed; Not counted for CGPA CA - Continuous Assessment FE-Final Examination

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S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER VII										
THEORY										
1	23U701	Logic and Distributed Control Systems	3	0	0	3	40	60	100	PC
2	23U702	Industrial Internet of Things	3	0	0	3	40	60	100	PC
3	23U___	Professional Elective III	3	0	0	3	40	60	100	PE
4	23U___	Professional Elective IV	3	0	0	3	40	60	100	PE
5	23____	Open Elective II	3	0	0	3	40	60	100	OE
PRACTICALS										
6	23U710	Industrial Automation Laboratory	0	0	4	2	60	40	100	EEC
7	23U711	Industrial Internet of Things Laboratory	0	0	4	2	60	40	100	EEC
8	23U720	Project Work I	0	0	4	2	60	40	100	EEC
		Total 27 hrs	**15**	**0**	**12**	**21**	**380**	**420**	**800**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER VIII										
THEORY										
1	23U___	Professional Elective V	3	0	0	3	40	60	100	PE
2	23U___	Professional Elective VI	3	0	0	3	40	60	100	PE
3	23U820	Project Work II	0	0	8	4	60	40	100	EEC
		Total 14 hrs	**6**	**0**	**8**	**10**	**140**	**160**	**300**	

** As per norms
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PROFESSIONAL ELECTIVES

Course Code	Course Title
23U001	Biomedical Instrumentation
23U002	Virtual Instrumentation
23U003	Fiber Optics and Laser Instrumentation
23U004	Power Plant Instrumentation
23U005	Industrial Chemical Processes
23U006	Thermodynamics and Fluid Mechanics
23U007	Instrumentation and Control in Petro Chemical Industries
23U008	Power Electronics and Drives
23U009	Robotics and Automation
23U010	Fundamentals of Pneumatics and Hydraulics
23U011	VLSI Design
23U012	Digital Image Processing
23U013	Computer Architecture
23U014	Operating Systems
23U015	Safety Instrumented Systems
23U016	Product Design and Development
23U017	Advanced Digital Signal Processing
23U018	Analytical Instrumentation
23U019	Smart Sensors and Actuators
23U020	Electric Vehicle System Modeling and Control
23U021	Computer Networks
23U022	Digital Twin
23U023	Instrumentation for Flowsheets
23U024	Industrial Cybersecurity

Professional Electives for BE Honours / BE Honours with specialization in same discipline and BE Minor degree programmes

VERTICAL I: CONTROL AND INDUSTRIAL AUTOMATION

23U031	Optimal and Adaptive Control Systems
23U032	Non-Linear Systems Theory
23U033	System Identification
23U034	Robust Control
23U035	Solid State Drives
23U036	Fault Diagnosis and Control
23U037	Model Based Control

VERTICAL II: SENSORS AND INSTRUMENTATION

23U041	State Estimation
23U042	Micro Electro Mechanical Systems
23U043	Multisensor Data Fusion
23U044	Advanced Sensing Techniques
23U045	Medical Imaging Systems
23U046	Sensor Systems Design

VERTICAL III: SIGNAL ANALYSIS

23U051	Wavelets and their Applications
23U052	Pattern recognition and Classification
23U053	Deep Learning for Signal Analysis
23U054	Biomedical Signal Processing
23U055	Medical Image Processing
23U056	Machine Vision

LANGUAGE ELECTIVES

23G001	Communication Skills for Engineers
23G002	Basic German
23G003	Basic French
23G004	Basic Japanese

OPEN ELECTIVES

AUTOMOBILE ENGINEERING

23AO01	Solar Vehicles
23AO02	Computational Fluid Dynamics
23AO03	Mobility and Infrastructure
23AO04	Artificial Intelligence and Machine Learning
23AO05	Flexible Manufacturing Systems

BIOMEDICAL ENGINEERING

23DO01	Nanophotonics
23DO02	Healthcare Information Systems

23DO03 Virtual Reality and Augmented Reality
23DO04 Biometric Systems
23DO05 Security for Medical Devices

BIOTECHNOLOGY
23BO01 Sports and Exercise Biochemistry
23BO02 Smart Farming and Precision Agriculture
23BO03 Water Resource Management
23BO04 Forensic Science

CIVIL ENGINEERING
23CO01 Statistics for Engineering Applications
23CO02 Optimization Techniques
23CO03 Safety in Construction Industry
23CO04 Psychology for Engineers
23CO05 Interior Design
23CO06 Urban Planning

COMPUTER SCIENCE AND ENGINEERING
23ZO01 Bioinformatics
23ZO02 Design Thinking
23ZO03 Intellectual Property Rights
23ZO04 Social and Economic Network Analysis

COMPUTER SCIENCE AND ENGINEERING (AI&ML)
23NO01 Bioinformatics
23NO02 Design Thinking
23NO03 Ethics of Artificial Intelligent
23NO04 Graph Theory
23NO05 Intellectual Property Right
23NO06 Social and Economic Network Analysis

ELECTRICAL AND ELECTRONICS ENGINEERING (Reg &SW)
23EO01 Electrical Safety Practices and Standards
23EO02 Design Thinking
23EO03 Waste to Energy
23EO04 Biology for Engineers
23EO05 Professional Ethics and Human Values

ELECTRONICS AND COMMUNICTION ENGINEERING
23LO01 Nano Technology and its applications
23LO02 Internet of things
23LO03 Virtual Reality
23LO04 Systems Engineering
23LO05 Universal Human Values 2: Understanding Harmony
23LO06 Artificial Intelligence and Machine Learning
23LO07 Measurements and Instrumentation

FASHION TECHNOLOGY
23HO01 Appreciation of Indian Art
23HO02 Creativity, innovation and design thinking
23HO03 Strategic management
23HO04 Human resource management
23HO05 Design Process for Social Innovation

INFORMATION TECHNOLOGY
23IO01 Information Technology in Biological Science
23IO02 Storage Technologies
23IO03 Green Computing
23IO04 Enterprise Resource Planning
23IO05 Technologies for Sustainable Systems
23IO06 Information Technology Management
23IO07 Total Quality Assurance
23IO08 Web Design and Development
23IO09 Information Systems
23IO10 E-Commerce
23IO11 Digital Marketing

INSTRUMENTATION AND CONTROL ENGINEERING
23UO01 Final Control Elements
23UO02 Energy Harvesting Techniques
23UO03 Industrial Communication Protocols
23UO04 Non Conventional Energy Systems
23UO05 Biology for Engineers
23UO06 Introduction to Machine Learning

MECHANICAL ENGINEERING (Reg & SW)

23MO01	Six Sigma Project Methodology
23MO02	Intellectual Property Rights
23MO03	Energy Resources, Economics and Environment
23MO04	Entrepreneurship Essentials
23MO05	Business Analytics
23MO06	Design Thinking
23MO07	Foundation Skills in Integrated Product Development

METALLURGICAL ENGINEERING

23YO01	Archaeometallurgy
23YO02	Industrial Safety and Energy Management
23YO03	Operations Research
23YO04	Sustainable Manufacturing
23YO05	Machine Learning

PRODUCTION ENGINEERING (Reg & SW)

23PO01	Battery Technology
23PO02	Digital Marketing for Engineers
23PO03	Financial Management
23PO04	Industrial Internet of Things
23PO05	Innovation and Leadership Management
23PO06	Professional Ethics
23PO07	Sustainable Development Goals for Manufacturing Industries
23PO08	Virtual Reality Systems and Applications

ROBOTICS AND AUTOMATION

23RO01	Robot Safety and Standards
23RO02	Decision Making
23RO03	Creative Kinetics
23RO04	Modeling and Simulation
23RO05	Optimization Techniques
23RO06	Virtual Instrumentation Systems
23RO07	Renewable Energy Systems
23RO08	Autotronics
23RO09	Database Management System
23RO10	Software Project Management and Quality Assurance
23RO11	Internet Tools and Java Programming
23RO12	Robotic Process Automation
23RO13	Blockchain Technology
23RO14	Mobile Application Development
23RO15	Human Values and Professional Ethics

TEXTILE TECHNOLOGY

23TH01	Characterization of Fibres and Polymers
23TH02	Sound and Thermal Insulation Products and Characterization
23TH03	Technical Textiles in Engineering Applications
23TH04	Electro Active Textiles
23TH05	Filtration Products and Characterization
23TH06	Industrial Textiles
23TH07	Textiles and Fabric care
23TO01	Circular Design and Sustainability
23TO02	Waste Management and Pollution Control
23TO03	Entrepreneurship
23TO04	Psychology of Stress, Health and Well-being
23TO05	Colour in Design
23TO06	Design Paradigm
23TO07	System Design for Sustainability

ONE CREDIT COURSES

23UF01	Advanced Industrial Automation Systems
23UF02	System Design and Implementation
23UF03	Calibration Techniques
23UF04	Motion Control Systems
23UF05	Electrical Metrology
23UF06	Standard Practices for Power Plant Instrumentation
23UF07	Automotive Instrumentation and Control
23UF08	Aircraft Instrumentation
23UF09	Automatic Flight Control System

HUMANITIES

23OFH01	Closed Loop Economy and Engineering Solutions

MATHEMATICS

23OFM01	Introduction to Quaternion Algebra

PHYSICS

23OFP01	Photonics
23OFP02	Lasers
23OFP03	Optics
23OFP04	Fiber optics
23OFP05	Quantum Photonics

CHEMISTRY

23OFC01	Battery Thermal Management System
23OFC02	Hydrogen Energy for Electrical Mobility

SELF DIRECTED LEARNING COURSES

23IC01	Experiencing Entrepreneurship
23IC02	Venture Development
23TL01	Public Administration
23TL02	Public Policy and Indian Economy

Summary of Credit Distribution

BE INSTRUMENTATION AND CONTROL ENGINEERING										
S. No	**Course Category**	**Credits Per Semester**								**Total Credits**
		I	**II**	**III**	**IV**	**V**	**VI**	**VII**	**VIII**	
1	HS	4	2	5	1	0	0	0	0	12
2	BS	7	9	8	4	0	0	0	0	28
3	ES	11	9	4	4	0	0	0	0	28
4	PC	0	0	6	15	17	15	6	0	59
5	PE	0	0	0	0	3	3	6	6	18
6	OE	0	0	0	0	0	3	3	0	6
7	EEC	0	0	1	1	1	2	6	4	15
8	MC	-	-	-	-	-	-	-	-	-
	TOTAL	**22**	**20**	**24**	**25**	**21**	**23**	**21**	**10**	**166**

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